

ASIS103 Foundations of Deep Learning

Midterm test, 1 hour 40 min, closed book

1. Multiple choices: choose one of the four answers.

1. In biological neurons, the typical time scale for synaptic transmission from one neuron to another is on the order of

- A. nanoseconds.
- B. microseconds.
- C. milliseconds.
- D. seconds.

2. Given the following PyTorch statements,

```
W = torch.randn(12)
```

```
V = W.reshape(-1,3)
```

The tensor V is a

- A. 1 by 3 random matrix.
- B. 4 by 3 uniform random matrix.
- C. -1 by 3 random matrix.
- D. 4 by 3 Gaussian random matrix.

3. Consider the following PyTorch code

```
import torch
x = torch.tensor([1., 2.], requires_grad = True)
y = x**2
z = (y*x).sum()
z.backward()
print(x.grad)
```

- A. tensor([1., 4.])
- B. tensor([2., 4.])
- C. tensor([3., 12.])
- D. tensor([3., 6.])

4. Let $\mathbf{y} = W\mathbf{x}$, $E = E(\mathbf{y})$, where W is an N by M matrix, $\mathbf{x}$, $\mathbf{y}$ are column vectors, and E is a scalar. Define the column gradient $\mathbf{g} = \frac{\partial E}{\partial \mathbf{y}}$. Which expression is correct for $\partial E / \partial W$?

- A. $\mathbf{g}\mathbf{x}^T$
- B. $\mathbf{x}\mathbf{g}^T$
- C. $\mathbf{g}W\mathbf{x}^T$
- D. $\mathbf{g}^T\mathbf{x}$

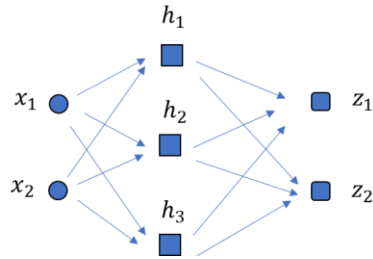
5. Consider the Kullback-Leibler (KL) divergence of distribution P with respect to Q , $D(P||Q)$. Which of the following statements is true?
 - A. The KL divergence is the average of $\log \frac{P}{Q}$ with respect to P .
 - B. The KL divergence is always strictly positive.
 - C. The KL divergence is the entropy minus the cross-entropy.
 - D. The KL divergence is the average of $\log \frac{P}{Q}$ with respect to Q .

6. A neural network gives a large training error and a large test error. After increasing the complexity of the network, the training error decreases substantially and the test error also decreases. Which statement best describes this observation?
 - A. The original model likely had high bias, and increasing the model complexity reduced the bias.
 - B. The original model likely had high variance, and increasing the model complexity reduced the variance.
 - C. The original model was overfitting, and increasing the model complexity reduced the overfitting.
 - D. Increasing model complexity always improves generalization.

7. The “no free lunch” theorem in machine learning implies that
 - A. no machine learning algorithm can perform better than random guessing.
 - B. a sufficiently large neural network can solve any learning problem better than other methods.
 - C. averaged over all possible learning problems, no learning algorithm is universally better than all others.
 - D. different learning algorithms give the same result when trained on the same data.

8. Given a feedforward multi-layer neural network with a total of P parameters, which of the following statements is false?
 - A. The computational complexity of a forward pass is $O(P)$.
 - B. Calculating the gradient using forward differences has computational complexity $O(P^2)$
 - C. The gradient calculation with reverse-mode automatic differentiation takes $O(P)$.
 - D. Calculating the gradient using central differences takes $O(P)$.

2. Consider a classification problem with two input features (x_1, x_2) and two output logits (z_1, z_2). The neural network has a hidden layer of three units with the tanh activation function, as shown below. The input data and class labels are generated by the following code.



- Define the 2-3-2 model with PyTorch. How many learnable parameters does this network have?
- Define the loss function using `nn.CrossEntropyLoss()` and the optimization routine with `torch.optim.SGD()` with a learning rate of `lr = 0.1`.
- Complete the training loop below for 200 epochs. In each epoch, perform a forward pass, compute the loss, perform backpropagation, and update the model parameters.

Code fragment:

```
import torch
import torch.nn as nn

X = 2*torch.rand(200,2) - 1
Y = (X[:,1] > X[:,0]**2 - 0.5).long()
# Y contains integer class labels

model = ...
loss_fn = ...
...

for epoch in range(200):
    # complete the training loop
    ...
```

3. An experimentalist finds that the result $t \in R$, for a given input vector $\mathbf{x}$, follows the probability distribution

$$p(t|\mathbf{x}) \propto e^{-|t-\boldsymbol{\theta}^T \mathbf{x}|},$$

where $\boldsymbol{\theta}$ is the model parameter vector.

- Determine the proportionality constant in the above equation so that the probability density is normalized to 1.

b) A data set has been collected with the pairs $(t_n, \mathbf{x}_n)$, $n = 1, 2, \dots, N$. Assuming that the data points are independent, derive the expression for the likelihood, and then obtain the negative log-likelihood (NLL).

c) Based on the result in part b), what error or loss function $E(\boldsymbol{\theta})$ should be used? Compute the gradient $\partial E / \partial \boldsymbol{\theta}$ analytically.

4. Consider an error function $E(w) = 2w + \frac{1}{2}w^2$ in a one-dimensional parameter space.

a) Starting from the position $w^{(0)} = 0$, use gradient descent with learning rate $\eta = 0.1$. What is the next position $w^{(1)}$?

b) Where is the exact minimum w^* ? With a learning rate of $\eta = 0.1$, roughly how many steps are needed to reach the minimum within an error tolerance $|w^{(k)} - w^*| < 10^{-3}$, starting from $w^{(0)} = 0$?

c) What range of the learning rate η gives convergence?

d) Now consider the two-dimensional error function $E(w_1, w_2) = 2w_1 + \frac{1}{2}w_1^2 + 2w_2^2$, where is the minimum? What range of η gives convergence for arbitrary starting points?

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